140. In a single throw of a pair of dice, what is the probability of getting a sum of 8?

(a) 8/12 (b) 1/36

(c) 4/36 (d) 5/36

Ans. (d) : In a single throw of a pair of dice, the probability of getting a sum of 8 is –

Total number of outcome = 6 × 6 = 36

Total possible outcome = (2, 6), (4, 4), (3, 5), (5, 3), (6, 2)

= 5

So the probability of getting a sum of 8 is

$$= \frac{5}{36}$$

141. A lottery system has ticket numbers with eight digits. There are ten first prizes of Rs. 1.00 lakh each, one hundred second prizes of Rs. 10,000 each and one thousand third prizes of Rs. 1,000 each.

At the draw of lots, an eight digit number is generated by choosing for each digit from left to right a random number in the range of 0 to 9.

Any one who has a ticket with the first two digits (from left to right) matching, the chosen number gets a first prize, and so on.

One individual purchases a large number of tickets at a discounted price of Re. 0.75 each. For how much money should he purchase tickets to achieve a 10% chance of winning a third prize?

(a) Rs. 10,000 (b) Rs. 75,000

(c) Rs. 1,00,000 (d) Rs. 7,500

Ans. (d) : To win a lottery of third prize of 1000 Rupees with a change of 10% at a discounted price of rupees 0.75 each. He should purchase a lottery of atleast 0.75 × 1000 × 100 = 7500 Rupees

142. The average salary of skilled employees in a firm is Rs. 520 and that of the unskilled employees is Rs. 420. If the average salary of both the groups of employees is Rs. 500, then the percentage of skilled employees in the firm is:

(a) 50 (b) 20

(c) 80 (d) 70

Ans. (c) : Let the skilled employee number is = x

Number of unskilled employee = y

Now,

$$520 \times x + 420\,xy = 500\,x + 500\,y$$

$$20x = 80y$$

$$x = 4y \quad \ldots\ldots\ldots(i)$$

Now the percentage of skilled employee is

$$= \frac{x}{x+y} \times 100 = \frac{4y}{4y+y} \times = 100\frac{4y}{5y} \times 100$$

= 80%

143. Consider the following pie charts?



The pie charts given here depict the area under cultivation of different cash crops, for two years, 1985 and 1990. Which one of the following statements is FALSE?

(a) In 1985, the area under jute cultivation was less than that under rice cultivation.

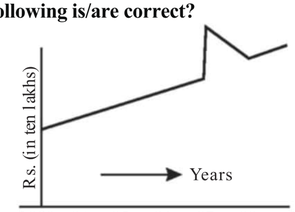
(b) More land was used for growing vegetables in 1990 than in 1985.

(c) In 1990, the area under rice and wheat cultivation taken together was less than half the total cultivation area.

(d) The percentage area under Jute cultivation was less in 1985 than in 1990.

Ans. (d) : From Pie graph its clear that, option (d) is incorrect as the percentage area under jute cultivation was more in 1985 as compared to 1990.

144. The following graph shows the annual sales of a company's products in millions of rupees. Study the trend and find out which of the following is/are correct?



I. A deliberate attempt has been made to distort the figures in one year.

II. Some one is interested in misleading the management.

III. The kink in the graph is unexplained.

Choose the correct answer from the codes given below:

(a) I only (b) II only

(c) III only (d) I and III

Ans. (c) : The kink in the graph is unexplained and exact explanation is not given to study deflection in the graph.

145. The following table shows the percentage change in the consumption of electricity by five towns P, Q, R, S and T from 1989 to 1991. Per cent change from 1989 to 1990 From 1990 to 1991:

Town	Percentage change	
	From 1989 to 1990	From 1990 to 1991
P	+8	18
Q	15	+11
R	+6	+9
S	-7	-5
T	+13	6

What was the percentage by which the total consumption of electricity by the five cities changed between 1989 and 1991?

(a) 4

(b) 4.45

(c) 4

(d) Data given are insufficient to answer

Ans. (d) : The total consumption of electricity by the five cities changed between 1989 and 1991 is not clear from the given table as data given are insufficient to answer.

146. The monthly incomes of Peter and Paul are in the ratio of 4:3. Their expenses are in the ratio of 3 : 2. If each saves Rs. 600 at the end of the month, their monthly incomes respectively are (in rupees)

(a) 3200 and 2400 (b) 2800 and 2100

(c) 2400 and 1800 (d) 2000 and 1500

Ans. (c) : Let Peter and Paul income be $4x$ and $3x$ and they both save 600 rupees at the end of the month and their expenses are in ratio 3 : 2.

$$\Rightarrow \frac{4x-600}{3x-600} = \frac{3}{2}$$

$$\Rightarrow 8x - 1200 = 9x - 1800$$

$$x = 600$$

Peter income is Rs. 2400 and Paul income is Rs. 1800.

147. If ABRA CADABRA is 1, 2, 9, 1 3, 1, 4, 1, 2, 9, 1, HOCUS POCUS will be:

(a) 5, 6, 3, 7, 8, 16, 15, 5, 21, 19

(b) 8, 16, 5, 21, 19, 16, 15, 5, 21, 19

(c) 8, 16, 5, 21, 19, 15, 16, 5, 21, 19

(d) 5, 6, 3, 7, 8, 6, 5, 3, 8, 7

Ans. (a) : According to question

A B R A ⟶ C A D A B R A

1 2 9 1 ⟶ 3 1 4 1 2 9 1

[R is 9th from the last]

Similary

H O C U S ⟶ P O C U S

Option (b) and (c) are wrong as given number are wrong.

where as

U S ⟶ 7,8

But in option (d) it is mentioned as 7, 8 and 8, 7 so (d) is also wrong.

So, only option (a) is correct.

148. In a group of six women there are four dancers, four vocal musicians, one actress and three violinists. Girija and Vanaja are among the violinists while Jalaja and Shailaja do not know how to play on the violin. Shailaja and Tanuja are among the dancers.

Jalaja, Vanaja, Shailaja and Tanuja are all vocal musicians and two of them are also violinists. If Pooja is an actress who among the following is both a dancer and a violinists?

(a) Jalaja (b) Shailaja

(c) Tanuja (d) Pooja

Ans. (c) : According to question in a group of six women there are four dancers, four vocal musicians one actress and three violinists.

But from statement it is clear that Girija and Vanaja are among the violinists and Jalaja and Shailaja do not known how to play violin.

Shailaja and Tanuja are dancer, so its very clear that Tanuja is the person who is both a dancer and a violinst.

149. If you are the ninth person in the queue starting from either end, than the number of persons in the queue is:

(a) 10 (b) 15

(c) 17 (d) 19

Ans. (c) : If the person is sitting at the 9th place from both the end so it is at middle position.

So, the number of person in queue is–

8 + 1 + 8 = 17

150. A father is now three times as old as his son. Five years back, he was four times as old as his son. The age of the son is:

(a) 12 years (b) 15 years

(c) 18 years (d) 20 years

Ans. (b) : Let the present age of father is x years.

and son's age is $\frac{x}{3}$ years

According to question

$$(x-5) = 4\left(\frac{x}{3} - 5\right)$$

$$\frac{x-5}{4} = \frac{x-15}{3}$$

$$3x - 15 = 4x - 60$$

$$\boxed{x = 45} \text{ years}$$

Present age of son is $= \frac{x}{3}$

$$= \frac{45}{3} = 15 \text{ years}$$